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Impact of Medication Therapy on Social, Behavioral, and Employment Outcomes in Adults With ADHD

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Abstract

Objective: Limited studies have explored the association between pharmacotherapy treatment and socio-behavioral and employment-related difficulties among adults with ADHD in U.S. This study aimed to evaluate and compare socio-behavioral and employment outcomes among adults with ADHD who were receiving stimulant treatment versus those who were not.

Methods: A retrospective cross-sectional study was conducted using the Medical Expenditure Panel Survey data from 2013-2019. Adults with ADHD were categorized into stimulant-treated versus not stimulant-treated groups. Multivariable logistic regression was utilized to assess the association between stimulant use and outcomes of interest (e.g., social, behavioral, and employment outcomes) and was reported using odds ratio (ORs). Propensity score matching was used to control for confounding and sub-group analyses were conducted by adherence to CNS stimulants.

Results: Among 1,290 (2,468,186 weighted) patients, 80.67% were stimulant-treated, and 19.33% were not treated with stimulants. No significant differences were found in the employment outcomes between the two groups in adjusted models. Sub-group analyses showed that adherent users of stimulants had higher odds of self-employment (OR = 2.10 [1.15, 3.82]) than non-adherent users. Stimulant-treated adults had higher odds of social limitations (OR = 3.16 [1.50, 6.69]), and lower odds of physical activity (OR = 0.29 [0.13, 0.66]) compared to those who were not treated with stimulants.

Conclusion: Overall, these findings highlight important gaps in understanding the relationship between real-world functional outcomes and stimulants use in adults with ADHD and underscore the need for additional longitudinal and real-world studies to thoroughly understand the impact of ADHD medications on those outcomes. (*J. of Att. Dis. 2026; 30(7) 916-929*)

Keywords
adult ADHD, CNS stimulants, socio-behavioral, employment

Introduction

Attention Deficit Hyperactivity Disorder (ADHD) is a neuro-psychiatric condition indicated by inattention, hyperactivity, and impulsivity that commonly occurs in children but can continue into adulthood (Faraone et al., 2024; Magnus et al., 2023; Wilens & Spencer, 2010). In 2018, it was estimated that 8.7 million adults were living with ADHD in the United States (U.S.; Schein et al., 2022). Among adults, it may often remain undiagnosed due to the misconception that it only affects children (Asherson et al., 2012; French et al., 2023; Ginsberg et al., 2014). Additionally, its symptoms frequently overlap with other conditions, such as anxiety, depression, and mood disorders, further complicating its diagnosis (Pallanti & Salerno, 2020). The Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5), published by the American Psychiatric Association (APA), mentions the criteria for diagnosing ADHD in adults (Cabral et al., 2020; Young & Goodman, 2016). In adults, ADHD diagnosis is confirmed if five or more symptoms of inattention and/or hyperactivity-impulsivity persist for at least 6 months (Cabral et al., 2020; Young & Goodman, 2016).

Adults with ADHD experience significant behavioral, economic, and humanistic challenges; for instance, ADHD-attributable total societal cost in adults was estimated at $122.8 billion in 2018 in the U.S., with 54.4% of these costs arising from unemployment, 23.4% from productivity loss, and 11.6% from healthcare service use (Schein et al., 2022).

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Furthermore, adults with ADHD are at an increased risk of engaging in risky behaviors, difficulty in work leading to reduced performance, difficulty in managing conflicts, and difficulty in maintaining personal and professional relations that contribute toward reduced quality of life (Björk et al., 2018; French et al., 2023; Kosheleff et al., 2023; Pawaskar et al., 2020).

The APA and the American Academy of Family Physicians guidelines recognized pharmacotherapy, particularly Central Nervous System (CNS) stimulants (e.g., dextroamphetamine and methylphenidate) and atomoxetine as first-line treatments for adults with ADHD, with antidepressants such as bupropion as alternatives to them (Post & Kurlansik, 2012). Extended-release formulations of CNS stimulants are preferred over immediate-release formulations because of the reduced dose intake. The reduction in dose frequency increases adherence to these medications. Cognitive-behavioral therapy is recommended concurrently with pharmacotherapy to improve management of the ADHD symptoms (Post & Kurlansik, 2012). Existing literature shows stimulants' effectiveness in enhancing daily functioning and quality of life in adults with ADHD (Coghill et al., 2017; Frampton, 2016; Surman et al., 2013; van Stralen et al., 2023). A systematic review and meta-analysis including several randomized controlled trials (RCTs), highlighted significant improvements in health-related quality of life and reduction in functional impairment among adults taking stimulants compared to placebo. For example, an effect size of at least 0.5 was reported for the stimulant group compared to the placebo group (Coghill et al., 2017). However, adherence to stimulants is essential for achieving therapeutic benefits in adult patients with ADHD.

Medication non-adherence results in a substantial economic burden, with an estimated annual cost of up to $177 billion, including both direct and indirect healthcare expenses (National Council on Patient Information and Education, 2007). Non-adherence to medication poses a significant risk, including disease progression and functional impairment, which can negatively impact social, behavioral, and employment outcomes in adults with ADHD (Kamimura-Nishimura et al., 2019; Kooij et al., 2013; Parkin et al., 2022). While stimulants effectively manage ADHD symptoms, they can also introduce side effects that can affect adherence (Kis et al., 2020; Knight, 2022; Nanda et al., 2023; Schein et al., 2023; Stevens et al., 2013). A recent study showed higher absenteeism among adults with low adherence to stimulant medications with a mean indirect cost of $3,669 (95% confidence interval [$2,396, $4,943]; Spalding et al., 2022). Furthermore, improved quality of life and work outcomes with CNS stimulants are associated with extended-release than immediate-release stimulants (Lee et al., 2023).

Despite the efficacy of stimulants in reducing symptom burden, their impact on broader social, behavioral, and employment outcomes among adults with ADHD is not well understood. There is limited evidence on the impact of stimulant use on social and behavioral outcomes among adults with ADHD in the real-world nationally representative U.S. populations. Most studies assessing functional and employment outcomes have been conducted in European populations (e.g., Sweden), which may not be applicable to the U.S. due to differences in patient population behaviors, social factors, and healthcare systems (Taipale et al., 2024). Furthermore, U.S.-based real-world studies have primarily focused on medication adherence, economic burden, and resource utilization, while a few studies have assessed socio-behavioral and employment outcomes (Jeun et al., 2024). Hence, it is important to conduct a population-based real-world U.S. study assessing the impact of stimulant use on socio-behavioral and employment factors among adults with ADHD. This will aid in providing a comprehensive understanding of the impact of pharmacotherapy on these outcomes beyond clinical trials and generalize findings to a diverse U.S. adult ADHD population.

To address these gaps, we conducted a cross-sectional study using a nationally representative survey data of U.S. adults. The overall aim of this study was to evaluate the impact of pharmacotherapy on social, behavioral, and employment outcomes in adults with ADHD. The primary aim was to evaluate and compare the social and behavioral outcomes in adults with ADHD treated with stimulants (stimulant-treated) compared to those who were not treated with stimulants (not stimulant-treated). We hypothesized that adults treated with stimulants would exhibit better social and behavioral outcomes compared to those who were not treated with stimulants. The secondary aim was to evaluate and compare employment outcomes between two groups. We hypothesized that adults treated with stimulants would exhibit favorable employment outcomes compared to who were not stimulant-treated.

Methods

Study Design

A retrospective cross-sectional study was conducted using the Medical Expenditure Panel Survey (MEPS) database, a nationally representative survey that provides information on respondent's demographic characteristics, health status, insurance coverage, access to care, employment status, prescriptions, patient satisfaction, and healthcare expenditures (Blewett et al., 2024).

Data Source

The MEPS database consists of two major components: household and insurance components. The Household Component (MEPS-HC) data are based on questionnaires asked to individual household members and their medical

providers. The Insurance Component (MEPS-IC) data come from a survey of employers conducted to collect health insurance plan information. We extracted the data from the full-year consolidated file, medical conditions file, prescription file, and the jobs file of the household component for the years 2013 to 2019. The full-year consolidated file provides information about demographics, income, person-level conditions, health status, disability days, quality of care, employment, health insurance, person-level medical care use, and expenditures. The medical conditions file contains information on the medical conditions reported by the individual household members and their providers. This information is eventually coded by professional coders to the International Classification of Diseases Clinical Modification 9th Revision (ICD-9-CM), or 10th Revision (ICD-10-CM) codes and included in the medical conditions data files to use for research (Agency of Healthcare Research and Quality, 2015b). The prescription file asks participants about their prescription drugs and their pharmacies to collect information on prescription start month, prescription start year, and prescription days of supply (Agency of Healthcare Research and Quality, 2015a). The jobs file collects information on a person's employment, including wages, hours, industry, and occupation.

Study Sample

The study included individuals with ADHD ICD-9-CM (314) and ICD-10-CM (F.90) and age $\geq$ 18 years without a cancer diagnosis. Cancer diagnoses were identified using the ICD-9-CM and ICD-10-CM codes (see Supplemental Table 1). For both aims, the study sample was categorized into stimulant-treated and not stimulant-treated based on whether the participant had a prescription for any of the four types of CNS stimulants, as identified by the National Drug Codes (NDC): Amphetamine/dextroamphetamine, Lis dexamphetamine, Methylphenidate, and Dexmethylphenidate (Supplemental Table 2). Individuals with missing information on NDC codes and drug names were excluded from the study. Not stimulant-treated group was defined as individuals with ADHD not taking any CNS simulants mentioned above. This group excluded individuals receiving non-stimulant medications, such as atomoxetine, viloxazine, clonidine, or guanfacine, and included only those who received no pharmacotherapy. Additionally, patients with no information in the jobs file were excluded from the dataset for the secondary aim. The detailed study sample selection process is depicted in Figure 1.

Dependent Variables

For the primary aim, five dependent variables related to social and behavioral difficulties were assessed: moderate/vigorous physical exercise five times per week, serious cognitive difficulties, difficulty doing errands alone, social limitations, and limitations in physical functioning. Four employment-related dependent variables were assessed for the secondary aim: employment status, self-employment status, paid sick leave, and paid vacation. Observations with missing values for the dependent variables were excluded from the study sample. All dependent variables were binary and identified from the full year consolidated and jobs files for each participant. The details of variable extraction from the survey questionnaire are described in Supplemental Table 3 (Agency of Healthcare Research and Quality, 2019).

Independent Variables

CNS stimulant use was the key independent binary variable. Other independent variables included were biological, socioeconomic, health-behavior-related, and comorbidities. Biological variables comprised of sex and age, with age categorized as 18 to 45, 46 to 65, and over 65 years. Socioeconomic variables included were race, ethnicity, region, marital status, education level, poverty level, and insurance status. Race was re-coded into a binary variable with White and non-White as the categories, while ethnicity was re-coded into Hispanic and non-Hispanic categories. Education level was re-coded as less than or equal to high school and more than high school. Smoking status was also recorded as a binary variable. Psychiatric comorbidities include depression, adjustment disorder, anxiety, bipolar disorder, conduct disorder, and personality disorder. Non-psychiatric comorbidities include heart disease, stroke, diabetes, arthritis, and chronic lung disease. The diagnosis for comorbidities was identified using the ICD-9-CM and ICD-10-CM codes (see Supplemental Table 1). Data on all these variables for each participant was obtained from the full-year consolidated file and medical conditions file for each year.

Statistical Analysis

Descriptive statistics were conducted to describe the patient's characteristics for both comparison groups: stimulant-treated versus not stimulant-treated. Mean and standard deviation were reported for the continuous variables, while frequency and percentage were reported for the categorical variables. Rao Scott's chi-square test was utilized to examine the differences in the categorical baseline characteristics between the two groups. *T*-test was used to compare the mean values between the groups. Multivariable logistic regression was conducted to demonstrate the relationship between the dependent and independent variables for primary and secondary aims. The odds ratio (OR) and *p*-value were reported for the unadjusted and adjusted models to show the association between dependent and independent variables. Since MEPS data uses a complex survey design, the weights associated with survey samples were accounted for in the analysis. These weights also adjust for the survey non-response (IPUMS MEPS, University of Minnesota, n.d.; Machlin et al., 2005). Only descriptives

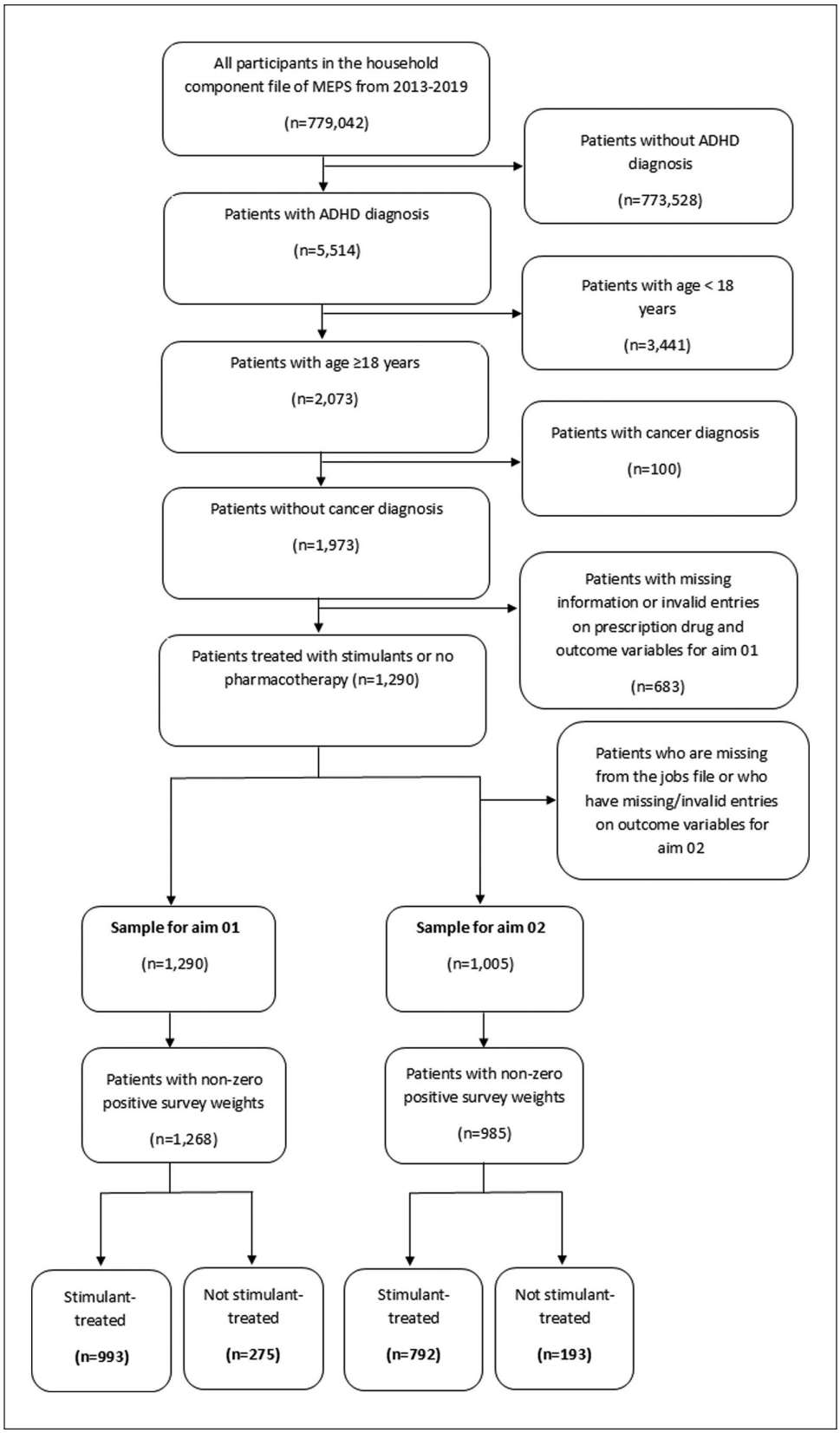


Figure 1. Flow diagram of sample selection process.

were reported for the participants with non-positive survey weights. Model diagnostics were assessed for the adjusted models, which showed a variance inflation factor (VIF) of less than five, indicating that there is no multicollinearity between the independent variables. The statistical significance was determined if p-value $\leq .05$. All the data management and analyses were conducted using SAS version 9.4 software (SAS Institute Inc., Cary, NC; SAS Institute Inc., 2013).

Subgroup Analysis

To assess the impact of medication adherence on social, behavioral, and employment variables, we calculated the medication adherence for participants treated with stimulants. The proportion of days covered (PDC) method was used to calculate adherence to stimulants (Forbes et al., 2018; Prieto-Merino et al., 2021). The number of days of medication supplied to the participant was divided by the number of days the participant was in the study period. The sample for each year was divided into new users and experienced users (Shah et al., 2020). New users were those who started the prescription in the year data was collected, while the experienced users started the prescription in the previous year(s). The number of days for experienced users was 365 or 366 (leap year) days. While, for the new year, the number of days was determined based on the month the participant was first prescribed the medication (Shah et al., 2020). Individuals with missing information on prescription drug supply days and year/month of medication start were excluded from the subgroup analysis. Based on the adherence threshold of $\geq$80%, participants were grouped as adherent stimulant users versus non-adherent stimulant users (Pharmacy Quality Alliance [PQA], 2022; Zhou et al., 2018)

Propensity Score Matching

To test the robustness of our findings, we employed propensity score matching (PSM) to ensure that the two comparison groups were balanced on relevant confounding variables and to minimize bias in estimating the effect of stimulant use on the outcomes (Austin, 2011). A 1:1 match was performed using the full matching technique without replacement (Austin, 2010). The propensity score model was developed using logistic regression, with CNS stimulant use as the outcome variable. Input variables in the PSM model included sex, age, race, insurance type, and number of non-psychiatric comorbidities. A generalized linear model (GLM) with binomial distribution and logit link function was used to estimate the probability of stimulant use based on these input variables. PSM was conducted in R statistical computing software version 4.2.3 (R Foundation for Statistical Computing, Vienna, Austria), employing the "MatchIt" and "optmatch" packages for PSM, "MASS" for statistical analysis, "cobalt" and "ggplot2" for visualizing covariate balance, and "survey" for analyzing complex survey data (i.e., MEPS; Greifer, 2024, 2025; Hansen & Klopfer, 2006; Ho et al., 2011; Lumley, 2004; R Core Team, 2023; Venables & Ripley, 2002; Wickham, 2016).

Results

The final study sample included 1,290 (2,468,186 weighted) patients who met the study inclusion criteria. The description of the baseline characteristics of the ADHD cohort by CNS stimulant use is presented in Table 1. About 80.67% (1,991,012) of the patients were stimulant-treated, and 19.33% (477,174) were not treated with stimulants. Based on weighted estimates, most of the patients in both groups were between the ages of 18 and 45 years (stimulant-treated vs. not stimulant-treated: 79.56% vs. 74.32%), were white (89.72% vs. 87.28%), not married (66.18% vs. 65.38%), had more than high school education (78.70% vs. 66.33%), were non-smokers (82.52% vs. 79.96%), and were residing in the Southern region (38.68% vs. 37.91%). Stimulant-treated group had a significantly greater number of people with high income than the not stimulant-treated group. (50.68% vs. 34.45%, $p < .001$). Although the majority of the patients in both groups had private insurance, the number was higher in the stimulant-treated group (82.30%) than in comparator group (65.94%). Only 7.07% in the stimulant-treated had greater than or equal to three psychiatric comorbidities compared to 15.64% in the not stimulant-treated group. The most prevalent psychiatric comorbidities in both groups were depression (25.01% vs. 37.62%, $p < .01$), anxiety (28.97% vs. 39.11%, $p < .05$), and bipolar disorder (11.82% vs. 24.02%, $p < .001$). Most of the patients in both groups had at least one non-psychiatric comorbidity (16.76% vs. 20.96%). Arthritis (16.78% vs. 26.87%, $p < .01$) and chronic lung disease (4.32% vs. 7.54%, $p = .06$) were the most prevalent non-psychiatric comorbidities. The percentage of the type of stimulant medications in the study cohort is depicted in Supplemental Figure 1. Amphetamine/dextroamphetamine (59.49%) was the most utilized stimulant medication in stimulant-treated group.

Table 2 presents the association between CNS stimulant use and dependent variables of the primary aim, considering both unadjusted and adjusted models. The adjusted model accounted for sex, age group, race, ethnicity, marital status, education, poverty level, insurance, and all comorbidities. In both the unadjusted and adjusted models, the OR for physical activity was not significantly associated with stimulant use. For serious cognitive difficulties, the unadjusted model showed no significant association with stimulant use. But the adjusted model showed that individuals treated with stimulants had significantly higher odds (OR = 1.47 [1.00, 2.15]; $p = .05$) of experiencing serious cognitive difficulties than those who were not treated with stimulants. Stimulant-treated adults had lower odds (OR = 0.43 [0.26, 0.71]; $p < .01$) of experiencing difficulty

Table 1. Description of the Sample by CNS Stimulant Use of Adults (≥18 years) With Attention Deficit Hyperactivity Disorder (ADHD), Who Participated in the Medical Expenditure Panel Survey (MEPS), 2013 to 2019.

Sample characteristics	Total population (W%)[a]	Stimulant treated[b]		Not stimulant-treated[c]		*p*-value[d]
		Weighted n	W %[a]	Weighted n	W %[a]	
Total population	2,468,186 (100)	1,991,012	80.67	477,174	19.33	
Sex						
Male	1,224,043 (49.59)	1,015,991	51.03	208,052	43.60	.08
Female	1,244,143 (50.41)	975,021	48.97	269,122	56.40	
Age group						
18–45 years	1,938,633 (78.54)	1,583,984	79.56	354,650	74.32	.17
46–65 years	464,099(18.80)	364,280	18.3	99,819	20.92	
>65 years	65,453 (2.65)	42,748	2.15	22,705	4.76	
Race						
White	2,203,463 (89.27)	1,786,321	89.72	417,142	87.42	.28
Non-White	264,723 (10.73)	204,691	10.28	60,032	12.58	
Ethnicity						
Hispanic	173,833 (7.04)	133,091	6.68	40,742	8.54	.28
Non-Hispanic	2,294,353 (92.96)	1,857,921	93.32	436,432	91.46	
Marital status						
Married	838,602 (33.98)	673,396	33.82	165,206	34.62	.85
Not married	1,629,584 (66.02)	1,317,616	66.18	311,968	65.38	
Region						
Northeast	413,411 (16.75)	341,157	17.14	72,254	15.14	.13
Midwest	672,281 (27.24)	563,534	28.3	108,747	22.79	
South	951,049 (38.53)	770,138	38.68	180,911	37.91	
West	431,445 (17.48)	316,183	15.88	115,262	24.16	
Education level[e]						
Less than or equal to high school	581,233 (23.70)	420,940	21.3	160,293	33.67	***
More than high school	1,870,866 (76.30)	1,555,157	78.7	315,708	66.33	
Poverty level						
Poor/low income	624,241 (25.29)	461,623	23.19	162,618	34.8	***
Middle income	670,465 (27.16)	520,312	26.13	150,153	31.47	
High income	1,173,480 (47.54)	1,009,077	50.68	164,403	34.45	
Insurance type						
Private	1,953,232 (79.14)	1,638,581	82.3	314,651	65.94	***
Public	434,807 (17.62)	293,696	14.75	141,111	29.57	
Uninsured	80,146 (3.25)	58,735	2.95	21,411	4.49	
Smoking status[e]						
Yes	407,613 (17.97)	321,475	17.48	86,138	20.04	.41
No	1,861,019 (82.03)	1,517,231	82.52	343,788	79.96	
Number of psychiatric comorbidities						
None	1,324,900 (53.68)	1,129,735	56.74	195,165	40.9	***
1	572,958 (23.21)	472,616	23.74	100,343	21.03	
2	354,933 (14.38)	247,909	12.45	107,024	22.43	
Greater than or equal to 3	215,394 (8.73)	140,753	7.07	74,642	15.64	
Depression						
Yes	677,526 (27.45)	498,033	25.01	179,493	37.62	**
No	1,790,660 (72.55)	1,492,979	74.99	297,681	62.38	
Adjustment disorder						
Yes	143,183 (5.80)	86,916	4.37	56,267	11.79	**
No	2,325,004 (94.20)	1,904,096	95.64	420,907	88.21	

(continued)

Table 1. (continued)

Sample characteristics	Total population (W%)[a]	Stimulant treated[b]		Not stimulant-treated[c]		*p*-value[d]
		Weighted n	W %[a]	Weighted n	W %[a]	
Anxiety						
Yes	763,403 (30.93)	576,764	28.97	186,640	39.11	*
No	1,704,783 (69.07)	1,414,248	71.03	290,534	60.89	
Bipolar disorder						
Yes	349,878 (14.18)	235,261	11.82	114,617	24.02	***
No	2,118,308 (85.82)	1,755,751	88.18	362,557	75.98	
Conduct disorder						
Yes	23,356 (0.95)	10,334	0.52	13,023	2.73	.1
No	2,444,830 (99.05)	1,980,679	99.48	464,151	97.27	
Personality disorder						
Yes	8,456 (0.34)	1,699	0.09	6,757	1.42	.05
No	2,459,730 (99.66)	1,989,313	99.92	470,417	98.58	
Number of non-psychiatric comorbidities						
None	1,867,917 (75.68)	1,547,126	77.71	320,791	67.23	*
1	433,721 (17.57)	333,697	16.76	100,024	20.96	
2	142,894 (5.79)	96,559	4.85	46,336	9.71	
Greater than or equal to 3	23.655 (0.96)	13,631	0.68	10,023	2.1	
Heart disease						
Yes	82,749 (3.35)	69,572	3.49	13,177	2.76	.5
No	2,385,437(96.65)	1,921,440	96.51	463,997	97.24	
Stroke						
Yes	18,310 (0.74)	6,599	0.33	11,711	2.45	.12
No	2,449,877 (99.26)	1,984,413	99.67	465,463	97.55	
Diabetes						
Yes	108,455 (4.39)	74,776	3.76	33,680	7.06	*
No	2,359,731 (95.61)	1,916,236	96.24	443,494	92.94	
Arthritis						
Yes	462,310 (18.73)	334,074	16.78	128,236	26.87	**
No	2,005,876 (81.27)	1,656,938	83.22	348,938	73.13	
Chronic lung disease						
Yes	121,876 (4.94)	85,914	4.32	35,962	7.54	.06
No	2,346,310 (95.06)	1,905,098	95.69	441,212	92.46	
All comorbidities						
None	1,069,733 (43.34)	944,097	47.42	125,636	26.33	***
1	634,629 (25.71)	521,358	26.19	113,270	23.74	
2	399,626 (16.19)	271,498	13.64	128,127	26.85	
Greater than or equal to 3	364,199 (14.76)	254,058	12.76	110,141	23.08	

[a]W% = weighted percentage.
[b]Stimulant-treated = participant who had a prescription for any of the four types of CNS stimulants: amphetamine/dextroamphetamine, lis-dexamphetamine, methylphenidate, and dexmethylphenidate.
[c]Not stimulant-treated = patients who received no pharmacotherapy.
[d]Significant at $p < .05$; statistically significant at $\alpha = .05$.
[e]Numbers may not add up to total due to missing values—education level has 0.7% missing data, and smoking status has 8.1% missing data.
*$p < .05$. **$p < .01$. ***$p < .001$.

doing errands alone compared to those not treated with stimulants in unadjusted model, but this association was not significant in the adjusted model. The unadjusted and adjusted odds ratio for social limitations was not significant with stimulant use. Individuals treated with stimulants had 1.68 times more odds ([1.08, 2.63]; $p < .05$) of having physical functioning limitations than those who were not treated with stimulants in the adjusted model.

Table 2.[a] Unadjusted (UOR) and Adjusted[b] (AOR) Odds Ratios and 95% Confidence Intervals From Multivariable Logistic Regression Determining the Association of the CNS Stimulant Use With Social and Behavioral Variables in Adults (≥18 years) With Attention Deficit Hyperactivity Disorder (ADHD) Using the Medical Expenditure Panel Survey (MEPS), 2013 to 2019.

	Social and behavioral variables									
	Physical activity		Serious cognitive difficulties		Difficulty doing errands alone		Social limitations		Limitations in physical functioning	
	UOR	AOR	UOR	AOR	UOR	AOR	UOR	AOR	UOR	AOR
Model	[95% CI]	[95% CI]	[95% CI]	[95% CI]	[95% CI]	[95% CI]	[95% CI]	[95% CI]	[95% CI]	[95% CI]
CNS stimulant use										
No [Ref]	—	—	—	—	—	—	—	—	—	—
Yes	1.02 [0.74, 1.39]	0.94 [0.68, 1.30]	0.9 [0.63, 1.29]	1.47 [1.00, 2.15]*	0.43 [0.26, 0.71]**	0.87 [0.56, 1.36]	0.68 [0.42, 1.09]	1.29 [0.76, 2.18]	0.85 [0.57, 1.28]	1.68 [1.08, 2.63]*
Sex										
Male [Ref]	—	—	—	—	—	—	—	—	—	—
Female	0.61 [0.47, 0.80]***	0.66 [0.50, 0.86]**	1.22 [0.92, 1.60]	1.01 [0.74, 1.38]	2.33 [1.36, 3.99]**	2.4 [1.23, 4.68]*	1.88 [1.30, 2.72]**	1.6 [0.99, 2.60]	1.87 [1.25, 2.78]**	1.41 [0.97, 2.04]
Age group										
18–45 years [Ref]	—	—	—	—	—	—	—	—	—	—
46–65 years	0.69 [0.50, 0.95]*	0.89 [0.64, 1.24]	0.98 [0.72, 1.34]	0.78 [0.55, 1.12]	1.14 [0.61, 2.15]	1.05 [0.52, 2.10]	1.38 [0.87, 2.19]	1.12 [0.69, 1.81]	3.64 [2.45, 5.41]***	2.67 [1.69, 4.22]***
>65 years	0.77 [0.40, 1.49]	1.06 [0.55, 2.06]	0.74 [0.36, 1.55]	0.42 [0.21, 0.84]*	0.64 [0.18, 2.33]	0.4 [0.09, 1.74]	3.12 [1.36, 7.12]*	1.96 [0.79, 4.89]	7.39 [3.72, 14.66]***	4.58 [1.92, 10.94]**
Race										
White [Ref]	—	—	—	—	—	—	—	—	—	—
Non-White	0.75 [0.51 – 1.10]	0.72 [0.48, 1.08]	2.01 [1.31, 3.19]**	1.57 [0.97, 2.55]	1.82 [1.0, 3.34]*	1.13 [0.61, 2.08]	2.73 [1.72, 4.36]***	2.13 [1.22, 3.72]*	1.21 [0.74, 2.0]	0.91 [0.56, 1.47]
Ethnicity										
Non-Hispanic [Ref]	—	—	—	—	—	—	—	—	—	—
Hispanic	1.17 [0.79, 1.73]	1.12 [0.74, 1.71]	1.18 [0.79, 1.76]	0.85 [0.53, 1.36]	1.14 [0.64, 2.02]	0.71 [0.38, 1.32]	0.77 [0.40, 1.47]	0.5 [0.27, 0.93]*	0.97 [0.59, 1.59]	0.86 [0.48, 1.55]
Marital status										
Not married [Ref]	—	—	—	—	—	—	—	—	—	—
Married	0.72 [0.55, 0.94]*	0.8 [0.60, 1.08]	0.71 [0.52, 0.98]*	0.83 [0.57, 1.22]	0.54 [0.30, 0.99]*	0.91 [0.38, 2.17]	0.51 [0.32, 0.80]**	0.51 [0.30, 0.87]*	1.56 [1.06, 2.29]*	2.04 [1.28, 3.28]**
Education level										
Less than or equal to high school [Ref]	—	—	—	—	—	—	—	—	—	—
More than high school	0.62 [0.47, 0.83]**	0.71 [0.51, 0.99]*	0.82 [0.60, 1.11]	1.1 [0.78, 1.56]	0.32 [0.18, 0.57]***	0.36 [0.16, 0.82]*	0.9 [0.57, 1.44]	1.21 [0.72, 2.02]	1.35 [0.89, 2.05]	1.1 [0.68, 1.79]
Poverty level										
Poor/low income [Ref]	—	—	—	—	—	—	—	—	—	—
Middle income	0.89 [0.64, 1.24]	0.89 [0.63, 1.27]	0.57 [0.41, 0.80]**	0.85 [0.59, 1.23]	0.64 [0.40, 1.03]	1.47 [0.87, 2.48]	0.51 [0.34, 0.78]**	0.8 [0.51, 1.28]	0.46 [0.28, 0.74]**	0.49 [0.29, 0.84]*
High income	0.97 [0.70, 1.34]	0.93 [0.63, 1.37]	0.34 [0.23, 0.51]***	0.61 [0.39, 0.97]*	0.24 [0.09, 0.67]*	0.83 [0.28, 2.46]	0.31 [0.18, 0.51]***	0.7 [0.38, 1.29]	0.25 [0.17, 0.36]***	0.32 [0.17, 0.59]***
Insurance type										
Private [Ref]	—	—	—	—	—	—	—	—	—	—
Public	1.06 [0.81, 1.39]	0.98 [0.72, 1.35]	3.27 [2.40, 4.47]***	2.35 [1.64, 3.35]***	6.56 [3.49, 12.32]***	4.17 [2.43, 7.17]***	3.91 [2.56, 5.97]***	2.38 [1.56, 3.64]***	4.6 [3.12, 6.78]***	3.03 [1.92, 4.78]***
Uninsured	1.11 [0.58, 2.15]	0.95 [0.52, 1.73]	1.04 [0.58, 1.86]	0.8 [0.41, 1.56]	1.91 [0.83, 4.36]	1.24 [0.54, 2.86]	2.51 [1.0, 6.38]*	1.88 [0.62, 5.69]	1.99 [0.68, 5.84]	1.33 [0.42, 4.19]
All comorbidities										
None [Ref]	—	—	—	—	—	—	—	—	—	—
1	0.66 [0.48, 0.92]*	0.73 [0.52, 1.03]	1.57 [1.05, 2.35]*	1.57 [1.05, 2.35]*	1.34 [0.42, 4.22]	1.2 [0.35, 4.18]	1.03 [0.54, 1.97]	0.84 [0.38, 1.86]	2.49 [1.25, 4.93]*	1.54 [0.78, 3.05]
2	0.66 [0.44, 0.98]*	0.73 [0.49, 1.10]	2.88 [1.80, 4.59]***	2.98 [1.88, 4.74]***	3.59 [1.08, 11.91]*	3.13 [0.80, 12.20]	3.17 [1.52, 6.61]**	2.73 [1.25, 5.95]*	5.57 [3.05, 10.19]***	3.8 [2.14, 6.74]***
Greater than or equal to 3	0.56 [0.39, 0.81]**	0.65 [0.45, 0.96]*	4.73 [3.01, 7.43]***	4.49 [2.86, 7.05]***	6.3 [2.06, 19.29]**	4.54 [1.37, 15.05]*	8.69 [4.55, 16.58]***	6.4 [3.37, 12.16]***	15.84 [9.35, 26.85]***	8.35 [4.80, 14.54]***

[a]Participants with non-positive person weights were excluded.
[b]Model adjusted for sex, age group, race, ethnicity, marital status, education level, poverty level, insurance type, and all comorbidities.
*$p < .05$. **$p < .01$. ***$p < .001$.

Table 3 shows association between CNS stimulant use and the dependent variables of the secondary aim for both unadjusted and adjusted models. The model was adjusted for the same covariates as for the primary aim. The unadjusted model showed that adults treated with stimulants had higher odds (OR = 1.75 [1.14, 2.67]; $p < .05$) of being employed than those who were not treated with stimulants. But this association was not significant after adjusting for covariates. No significant association was found between self-employment status and key independent variable in both unadjusted and adjusted models. The unadjusted model for paid vacation showed that stimulant-treated group had significantly higher odds (OR = 1.95 [1.07, 3.55]; $p < 0.05$) of taking a paid vacation than those not taking stimulants. But this association was not significant in the adjusted model. Lastly, the odds ratio for paid sick leave was not significant in either of the models with CNS stimulant use.

Sub-Group Analysis

Supplemental Table 4 presents the baseline characteristics for the sub-group analysis conducted by adherence to CNS stimulants. Adherence was obtained for 637 patients (1,283,872 weighted). Based on weighted estimates, 65.57% were non-adherent to stimulants, while 34.43% were adherent users. Adherent users smoked more than non-adherent users (non-adherent vs. adherent: 17.62% vs. 24.43%). More number of adherent users had at least two psychiatric comorbidities compared to non-adherent users (7.79% vs. 17.07%; $p < 0.001$). The most prevalent psychiatric comorbidities in both the groups were anxiety (22.50% vs. 41.96%; $p < .001$), depression (19.63% vs. 31.06%; $p < .01$), and bipolar disorder (6.92% vs. 13.56%; $p < .01$). More adherent users had arthritis than non-adherent users (10.10% vs. 20.20%; $p < .01$).

Supplemental Tables 5 and 6 shows the association between CNS stimulant adherence and outcome variables among adults (≥18 years) with ADHD for the adjusted model. None of the ORs were significant for CNS stimulant adherence with the social and behavioral outcome variables. The adjusted model with self-employment status as the outcome variable showed that adherent users had 2.10 times higher odds of being self-employed than non-adherent users ([1.15, 3.82]; $p < .05$). No statistical inference was estimated for paid vacation and paid sick leave as all the observations had the same response.

Propensity Score Matched Cohorts

The sample distribution before and after matching is described in Supplemental Tables 7 and 8. The initial covariate imbalances between the stimulant-treated and not stimulant-treated groups were reduced after matching, resulting in no significant difference between the groups for sex, race, age, insurance, and non-psychiatric comorbidities. Overall, multivariable logistic regression with matched data showed a similar direction of association for the socio-behavioral and employment outcomes compared to the main (unmatched data) analysis. However, the results showed stimulant-treated adults had significantly lower odds of engaging in physical activity (OR = 0.29 [0.13, 0.66]; $p < .01$) and higher odds of experiencing social limitations (OR = 3.16 [1.50, 6.69]; $p < .01$) than those not treated with stimulants, outcomes which were insignificant in the main analysis. Moreover, in the matched analysis, stimulant use was not significantly associated with serious cognitive difficulties or physical functioning limitations, in contrast to the findings from the unmatched data. Supplemental Tables 9 and 10 present the odds ratios for the outcome variables in both aims for the matched and unmatched data.

Discussion

Our study objective was to investigate and compare the social, behavioral, and employment outcomes among adults with ADHD who were treated with stimulants compared to those who were not. We did not find any significant associations between the two groups for employment-related outcomes in adjusted model. However, the unadjusted model for employment status and paid vacation showed higher odds of employment and paid vacation for stimulant-treated adults compared to those who were not treated with stimulants (see Table 3 and Supplemental Table 10). Interestingly, the subgroup analysis showed that adherent users of stimulants had higher odds of being self-employed than non-adherent users (see supplemental Table 6). We also found that adults treated with stimulants have higher odds of experiencing social limitations and lower odds of engaging in physical activity compared to those not treated with stimulants (see Supplemental Table 9).

Existing evidence on the impact of stimulant use on employment outcomes is heterogeneous and limited. Most studies suggest that stimulant medication use may be associated with improved employment-related outcomes compared with no pharmacotherapy. However, the outcomes evaluated in these studies might differ from those assessed in the current study and were used as the most closely related evidence available. For example, a nationwide Swedish study conducted using both within-individual and between-individual designs found that stimulant use was not associated with reduced work disability when individuals were compared to nonuse in the within-individual analyses (Taipale et al., 2024). Whereas between-individual comparisons showed that stimulant use was associated with lower work disability compared to untreated individuals. Similarly, another Swedish registry study reported lower risk of long-term unemployment among middle-aged women (30-55 years) with ADHD medication use (stimulants and non-stimulants) compared to its non-use (Li et al., 2022). In contrast, a US-based study assessing work productivity and activity impairment reported no significant differences between stimulant treated and unmedicated adults with ADHD (Lee et al., 2023). Consistent with the mixed

Table 3[a]. Unadjusted (UOR) and Adjusted[b] (AOR) Odds Ratios and 95% Confidence Intervals From Multivariable Logistic Regression Determining the Association of the CNS Stimulant Use With Employment-related Variables in Adults (≥18 years) With Attention Deficit Hyperactivity Disorder (ADHD) Using the Medical Expenditure Panel Survey (MEPS), 2013 to 2019.

	Employment, related variables							
	Employment status		Self, employment status		Paid vacation		Paid sick leave	
	UOR	AOR	UOR	AOR	UOR	AOR	UOR	AOR
Model	[95% CI]	[95% CI]	[95% CI]	[95% CI]	[95% CI]	[95% CI]	[95% CI]	[95% CI]
CNS stimulant use								
No [Ref]	—	—	—	—	—	—	—	—
Yes	1.75 [1.14, 2.67]*	0.86 [0.51, 1.45]	0.8 [0.42, 1.52]	0.88 [0.43, 1.82]	1.95 [1.07, 3.55]*	1.71 [0.91, 3.21]	1.44 [0.79, 2.64]	1.32 [0.71, 2.48]
Sex								
Male [Ref]	—	—	—	—	—	—	—	—
Female	0.52 [0.35, 0.79]**	0.61 [0.37, 1.0]*	1 [0.62, 1.59]	0.98 [0.57, 1.69]	0.87 [0.49, 1.54]	0.75 [0.39, 1.45]	0.83 [0.47, 1.47]	0.71 [0.36, 1.40]
Age group								
18–45 years [Ref]	—	—	—	—	—	—	—	—
46–65 years	0.41 [0.28, 0.60]***	0.43 [0.27, 0.69]**	1.61 [0.93, 2.78]	1.1 [0.59, 2.04]	2.52 [1.64, 3.90]***	2.26 [1.44, 3.55]***	2.75 [1.73, 4.37]***	2.39 [1.48, 3.87]**
>65 years	0.07 [0.04, 0.14]***	0.06 [0.03, 0.14]***	1.5 [0.65, 3.44]	0.73 [0.25, 2.13]	0.87 [0.53, 1.44]	0.8 [0.43, 1.48]	1.62 [0.62, 4.21]	1.39 [0.50, 3.87]
Race								
White [Ref]	—	—	—	—	—	—	—	—
Non, White	0.74 [0.45, 1.22]	0.91 [0.51, 1.62]	0.41 [0.22, 0.77]**	0.48 [0.25, 0.94]*	0.92 [0.43, 1.95]	1.03 [0.48, 2.21]	0.73 [0.29, 1.84]	0.91 [0.36, 2.28]
Ethnicity								
Non, Hispanic [Ref]	—	—	—	—	—	—	—	—
Hispanic	1.8 [0.87, 3.75]	2.12 [0.98, 4.61]	0.4 [0.12, 1.34]	0.51 [0.14, 1.77]	0.84 [0.33, 2.11]	1.05 [0.43, 2.59]	0.79 [0.29, 2.10]	1.16 [0.43, 3.13]
Marital status								
Not married [Ref]	—	—	—	—	—	—	—	—
Married	0.97 [0.65, 1.44]	0.73 [0.44, 1.22]	2.6 [1.59, 4.26]***	2.23 [1.31, 3.81]**	1.31 [0.81, 2.12]	0.84 [0.52, 1.36]	1.78 [1.08, 2.93]*	1.05 [0.64, 1.73]
Education level								
Less than or equal to high school [Ref]	—	—	—	—	—	—	—	—
More than high school	2.01 [1.33, 3.02]**	2.72 [1.58, 4.66]***	1.84 [0.95, 3.56]	1.63 [0.82, 3.23]	12.13 [2.19, 67.14]**	8.07 [1.36, 47.98]*	11.65 [2.18, 62.34]**	7.31 [1.28, 41.78]*
Poverty level								
Poor/low income [Ref]	—	—	—	—	—	—	—	—
Middle income	2.06 [1.34, 3.17]**	1.58 [1.0, 2.50]*	1.08 [0.55, 2.10]	1.23 [0.61, 2.46]	1.32 [0.56, 3.10]	0.97 [0.38, 2.48]	2.29 [0.80, 6.51]	1.59 [0.54, 4.65]
High income	4.11 [2.78, 6.08]***	2.4 [1.27, 4.56]*	1.48 [0.76, 2.90]	2.15 [1.0, 4.67]*	1.71 [0.77, 3.80]	1.06 [0.43, 2.59]	3.10 [1.12, 8.57]*	1.76 [0.59, 5.25]
Insurance type								
Private [Ref]	—	—	—	—	—	—	—	—
Public	0.16 [0.10, 0.25]***	0.3 [0.16, 0.54]***	1.85 [1.07, 3.21]*	3.42 [1.83, 6.38]***	0 [0.00, 0.00]	0 [0.00, 0.00]	0 [0.00, 0.00]	0 [0.00, 0.00]
Uninsured	0.24 [0.11, 0.53]**	0.27 [0.10, 0.80]*	3.71 [1.68, 8.18]**	5.27 [2.13, 13.03]***	0 [0.00, 0.00]	0 [0.00, 0.00]	0 [0.00, 0.00]	0 [0.00, 0.00]
All comorbidities								
None [Ref]	—	—	—	—	—	—	—	—
1	0.38 [0.21, 0.68]**	0.49 [0.26, 0.91]*	1.5 [0.81, 2.77]	1.37 [0.70, 2.69]	1.45 [0.64, 3.29]	1.62 [0.68, 3.90]	1.58 [0.70, 3.60]	1.75 [0.73, 4.17]
2	0.28 [0.15, 0.53]***	0.44 [0.22, 0.86]*	1.1 [0.52, 2.32]	1.03 [0.45, 2.36]	1.1 [0.42, 2.85]	1.35 [0.46, 3.97]	1.3 [0.51, 3.32]	1.55 [0.53, 4.56]
Greater than or equal to 3	0.14 [0.08, 0.24]***	0.27 [0.15, 0.47]***	1.91 [1.05, 3.47]*	1.67 [0.78, 3.55]	1.99 [0.98, 4.02]	2.3 [0.98, 5.38]	1.76 [0.80, 3.84]	1.97 [0.78, 4.97]

[a]Participants with non-positive person weights were excluded.
[b]Model adjusted for sex, age group, race, ethnicity, marital status, education level, poverty level, insurance type, and all comorbidities.
*$p < .05$. **$p < .01$. ***$p < .001$.

literature and contrary to our study hypothesis, the results showed no significant associations between stimulant use and better employment outcomes.

Our unadjusted results indicated higher odds of employment and paid vacation among adults treated with stimulants compared to those who were not, suggesting better employment outcomes in the stimulant-treated group. However, this association did not stand out after adjusting for covariates, suggesting that the difference observed could be due to baseline group differences rather than treatment effects. These results align with the existing literature, which suggests that while stimulants reduce core symptoms (i.e., inattention and hyperactivity) of ADHD, which could interfere with work performance, their direct impact on the functional and employment outcomes is less consistent (Coghill et al., 2017; Taipale et al., 2024). In addition, multiple factors could impact the association between stimulant use and employment outcomes, leading to contradictory results from our hypothesis. For example, adults treated with stimulants may experience treatment side effects such as insomnia, irritability, anxiety, and mood swings, which could outweigh the potential therapeutic benefits of stimulants in improving employment outcomes. Additionally, confounding by indication could be another explanation for poor outcomes in the stimulant-treated group. For example, stimulants could be prescribed for adults with more severe ADHD symptoms (disease severity) and hence it could impact the study outcomes. Another factor that could impact our results is patient behavior, such as medication adherence, as shown by our subgroup analysis, which reported that adults who were adherent to stimulants had better employment outcomes than those who were not adherent to stimulants (see Supplemental Table 6). This can be further supported by a U.S.-based study, which reported that adults with ADHD who had low adherence to stimulants experienced significantly greater levels of absenteeism, work productivity loss, and activity impairment than those with high medication adherence (Spalding et al., 2022).

Furthermore, the literature on the association between stimulant use and social and behavioral outcomes in adults with ADHD is scarce. Studies have reported that adults with ADHD who are treated with stimulants have better health-related quality of life (QOL) and functional outcomes compared to placebo (Coghill et al., 2017; Kosheleff et al., 2023). For example, a randomized placebo-controlled trial of lisdexamfetamine (LDX) in adults with ADHD reported improvements in domains of performance and daily functioning (effect size [ES] = 0.93, $p < .0001$), general well-being (ES = 0.70, $p < .0001$), and relationships/communication (ES = 0.31, $p = .03$) compared to placebo (Adler et al., 2013). However, literature has also reported that although stimulants improve QOL and functional outcomes, their effects on these measures are smaller than those observed for relief of core ADHD symptoms (Coghill et al., 2017). Contrary to the study hypothesis, our findings suggest that stimulant-treated adults experience greater social limitations, as well as lower physical activity, which may be influenced by multiple factors. First, side effects of stimulants such as insomnia and irritability could negatively impact social and behavioral outcomes (Schein et al., 2023). Also, individuals receiving stimulants might have more disease severity than those who were not treated with stimulants, leading to confounding by indication. The baseline differences between the groups could also impact the study results. Although we conducted propensity score matching to balance observable baseline characteristics between the two groups, some unmeasured confounders such as ADHD disease severity and treatment side effects could also influence the study results.

Our study has several strengths. To our knowledge, this is the first study to assess the impact of stimulant use on social, behavioral, and employment outcomes in a sample of U.S. adults with ADHD. We utilized a national survey dataset over a long time period (2013–2019), which enhanced the generalizability of the findings to the entire U.S. population with ADHD. Additionally, we assessed the outcomes based on adherence to stimulants in the subgroup analysis. Also, the number of patients by medication type and adherence rates was validated with another study conducted in a similar ADHD population, suggesting robustness in our data management steps (Jeun et al., 2024). Lastly, we validated and confirmed the robustness of the results through propensity score matching.

Despite its strengths, our study is not without limitations. First, due to the cross-sectional study design, we were not able to assess changes in outcomes for the population over time. Second, the dataset lacked information on factors that could impact the results, such as ADHD severity, medication effectiveness, side effects, and childhood ADHD history (Halmøy et al., 2009; Kosheleff et al., 2023). A study showed that stimulant use during childhood is the strongest predictor of employment in adulthood (Halmøy et al., 2009). Information on the childhood history of ADHD would have provided a better understanding of our findings. Third, since most of the study population belonged to the White race, these results cannot be generalized to other racial and ethnic groups, as they might vary (Cummings et al., 2017). Fourth, self-reported data were obtained from the study participants, which could lead to information biases such as recall bias, interviewer bias, social desirability bias, and volunteer bias (Althubaiti, 2016). Fifth, the exclusion of approximately 900 patients with missing or incomplete information on prescription medication or employment-related information may have introduced selection bias. Sixth, participants taking non-stimulant medications or a combination of medication agents were not included, as the focus was on comparing stimulant-treated participants with those who were not treated with stimulants. Hence, the impact of multiple agents on the outcomes cannot be determined from this study. Lastly, due to data limitations, the use of non-pharmacological treatments such as behavioral therapy cannot be reliably identified or quantified and was not included in the study.

Our findings can aid clinicians in developing treatment plans by considering the impact of stimulant use on functional impairments and employment outcomes in adults with ADHD. Also, the increased awareness of the adverse effects of these medications can lead to better-informed individualized treatments. Additionally, the existing evidence, that is, randomized controlled trials and observational studies assessing social, behavioral, and employment outcomes for stimulants in the adult U.S. ADHD population is limited, as most of the studies are conducted in ADHD patients aged ≤18 years, making direct comparison of our study findings with the literature challenging. This suggests that more randomized trials and longitudinal studies are needed to explore the long-term effects of stimulant use and other ADHD medications on social, behavioral, and employment outcomes in adult ADHD. Future research is also needed to investigate how different subgroups (e.g., race and disease severity) respond to these treatments in a real-world setting, as this could improve our understanding of ADHD and lead to more tailored therapeutic approaches for the patients based on their needs.

In conclusion, adults with ADHD who were treated with stimulants had higher odds of social limitations and lower odds of engaging in physical activity than those who were not treated with stimulants. Although statistically insignificant, stimulant-treated adults demonstrated slightly better employment outcomes than those not treated with stimulants. These findings suggest that the association between social, behavioral, and employment outcomes and ADHD medication in adults is complex and multifaceted. We emphasize the need for more real-world data on ADHD pharmacotherapy and further clinical trials to understand these outcomes by medication use in adult ADHD to aid in the development of more effective treatment strategies for the patients.

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Data Availability Statement

The data that support the findings of this study are openly available at https://meps.ahrq.gov/mepsweb/data_stats/download_data_files.jsp.

Supplemental Material

Supplemental material for this article is available online.

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